

Demonstration of Full C-band 400G ZR Transmission over 3-Span 427.97-km Hollow-Core Fibre

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Abstract *We experimentally demonstrate 400G ZR transmission over record reach thanks to the ultra-low nonlinearity and low-and-flat chromatic dispersion of hollow-core fibre (HCF). Using 34.5-dBm launch powers, real-time full C-band 400G ZR transmission over 3 spans with a maximum 427.97-km reach is demonstrated. ©2026 The Author(s)*

Introduction

The rapid expansion of cloud/edge services and AI-driven workloads is pushing optical transmission systems towards high capacity with low latency, whilst maintaining low cost, low power consumption and operational simplicity. In this context, 400G ZR pluggable coherent optics have emerged as a key enabler of IP-over-DWDM architectures by integrating coherent transceivers directly into routers/switches [1, 2]. This integration streamlines network operations and contributes to reduced costs, power consumption, space requirements, as well as lowering operational complexity. As a result, 400G ZR optics have been widely deployed in data centre interconnects and metropolitan links [1, 2].

However, the economic appeal of 400G ZR is tightly coupled to its achievable reach, which is typically in the range of ~80 to 120 km for standard single-mode fibers (SMFs) [3-6]. This is partially limited by the achievable signal-to-noise ratio (OSNR) in the moderate launch power regime dictated by glass-fiber nonlinearity. Another key limit is the chromatic dispersion (CD) that can be practically handled within the power and thermal budgets of compact pluggable modules. This is capped at 2,400 ps/nm for the fibre link according to the OIF 400G ZR implementation agreement [7]. Extending the reaches typically demands higher-performance coherent optics such as the 'ZR+' variants [5, 6] but this results in increased costs and higher power consumption.

Hollow-core fibres (HCFs) provide a compelling route to extend ZR-class reach because they offer orders-of-magnitude lower fibre nonlinearity and low-and-flat CD over wide bandwidths, while also delivering inherent >30% latency reduction [8, 9]. The ultra-low nonlinearity of HCFs enables the use of high-power optical amplification to achieve higher launch powers and support significantly longer span lengths [10-18]. On the other hand, the low CD of HCF can directly relax the dispersion burden seen by the receiver DSP, improving the feasibility of longer reaches while remaining aligned with ZR-class CD tolerance and power constraints. The synergy between 400G

ZR optics and HCFs has been explored in a few prior works [18, 19], with up to 107.5-km reach demonstrated in [18] and the longest 400G ZR transmission over ultra-low loss SMF to date being 151 km [5].

In this paper, we investigate a 400G ZR-centric reach-extension concept that exploits the uniquely-low nonlinearity and CD of HCF to push 400G ZR optics further into extended-metro/regional distances while preserving the low-cost, energy-efficient pluggable paradigm. With the help of 34.5-dBm high-power (HP) amplifiers, full C-band 400G ZR transmission is demonstrated over single-span 163.9-km, 2-span 315.97-km and 3-span 427.97-km HCF links. These results represent record-length 400G ZR transmission to date and have significantly exceeded the reach limit that can be achieved in SMF. We believe this work showcases an unprecedented new route towards future cost-effective and power-efficient optical interconnects by exploring the use of 400G ZR optics beyond existing metro applications.

Experimental Setup

Fig. 1(a) shows the experimental setup of the 400G ZR transmission over up to 3-span HCF links. At the transmitter (Tx), a 400G ZR pluggable generating ~60-GBaud 16-ary quadrature amplitude modulation signal in real-time was mounted onto an optical network tester (ONT) to serve as the channel-under-test (CUT). The CUT's output power was kept at -10dBm. A WaveShaper (WS) was adopted to shape the output of an amplified spontaneous emission (ASE) source to generate the remaining 63×75-GHz-spaced dummy channels (also emulated as 60 GBaud). To assess the transmission performance across all 64 channels within the C-band, the central frequency of the 400G ZR CUT was varied from 191.375 THz to 196.1 THz, with adjustments made to the ASE dummy channels' spectra accordingly. The combined signal was then fed into the high-power (HP) booster amplifier (BA) to provide 34.5-dBm launch power into the fibre.

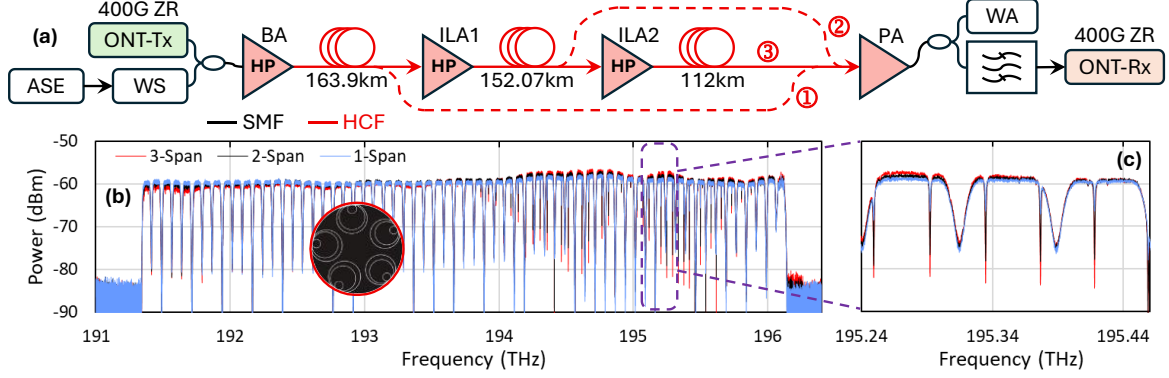


Fig. 1: (a) Setup of 400G ZR transmission over HCF links, (b) optical spectra after 1-span, 2-span and 3-span of HCF links at 150-MHz resolution, (c) zoom in of the optical spectra over 195.24 to 195.46 THz. Inset: cross section image of the HCF.

As shown in Fig. 1(a), we considered three different scenarios for performance evaluation: case ① refers to the single-span 163.9-km transmission scenario, whereas in cases ② and ③, two and three spans were adopted with the help of additional 34.5-dBm in-line amplifiers (ILAs), respectively. The lengths of the second and third spans were 152.07 km, and 112 km, respectively. The three spans incorporated 41 fiber bobbins, with lengths varying between ~ 1.82 km and 20.12 km. The average loss of these fibre bobbins ranged from 0.1 dB/km to 0.24 dB/km. Splicing was performed using a rotational alignment technique. The total span losses including splice and connector losses were around 32.4 dB, 30.5 dB, and 27.8 dB, respectively. The gas line absorption (GLA) factor, which refers to the line depth at the highest absorbing C-band frequency of around 195.33 THz, was moderately high with an average of ~ 0.05 dB/km due to the presence of some abnormally high GLA fibers needed to reach the desired span lengths. The HCF operated in the fundamental anti-resonant window, and its CD and mean polarization mode dispersion values in the C-band were around 4 ps/nm/km and 0.3 ps/ $\sqrt{\text{km}}$, respectively [11]. As illustrated in the inset of Fig. 1, all the fibres used in this work adopted the double-nested anti-

resonant nodeless design, which is comparable to the fibre structure detailed in [8]. The inter-modal interference of the fiber was measured to be better than -56 dB/km.

At the receiver (Rx), the signal was first amplified by a pre-amplifier (PA) with 23.5-dBm output power. A tunable filter was then utilized to select the CUT for real-time signal detection by the 400G ZR pluggable, which reports the pre-forward error correction bit error rate (pre-FEC BER) and the estimated equivalent OSNR by the module's versatile diagnostics monitoring. A 10% tap of the PA's output was captured by a WaveAnalyzer (WA) with a resolution of 150-MHz to monitor the optical spectra. The comparison of optical spectra after 1-span (163.9 km, case ①), 2-span (315.97 km, case ②) and 3-span (427.97 km, case ③) HCF links is given in Fig. 1(b). It is shown that the severity of GLA increases over longer distances of HCF, reaching more than 20 dB in some parts of the spectra. We note that the power profile fluctuations at different channels shown in Fig. 1(b) are mainly attributed to gain differences between the optical amplifiers.

Experimental Results

We first investigated the full C-band 400G ZR transmission over the single-span 163.9-km (i.e.,

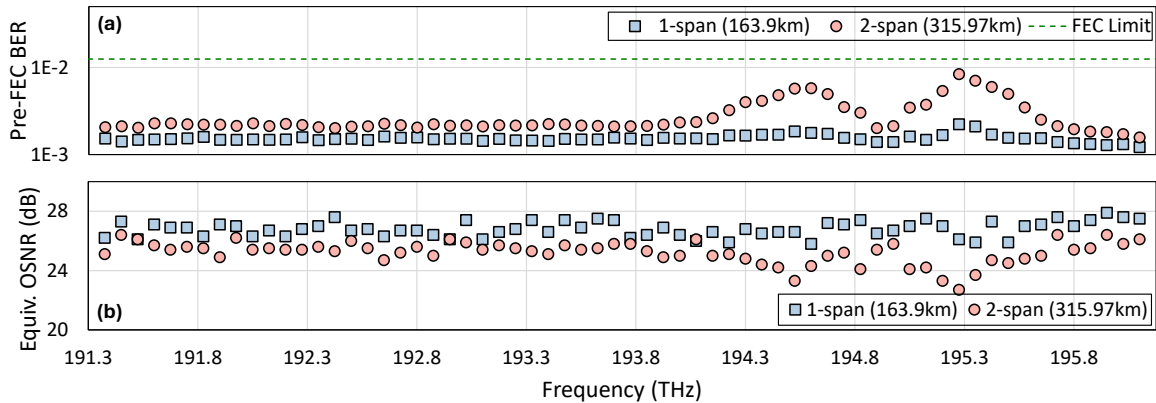


Fig. 2: Results of full C-band 400G ZR transmission over 1-span and 2-span HCF links: (a) pre-FEC BER, and (b) estimated equivalent OSNR.

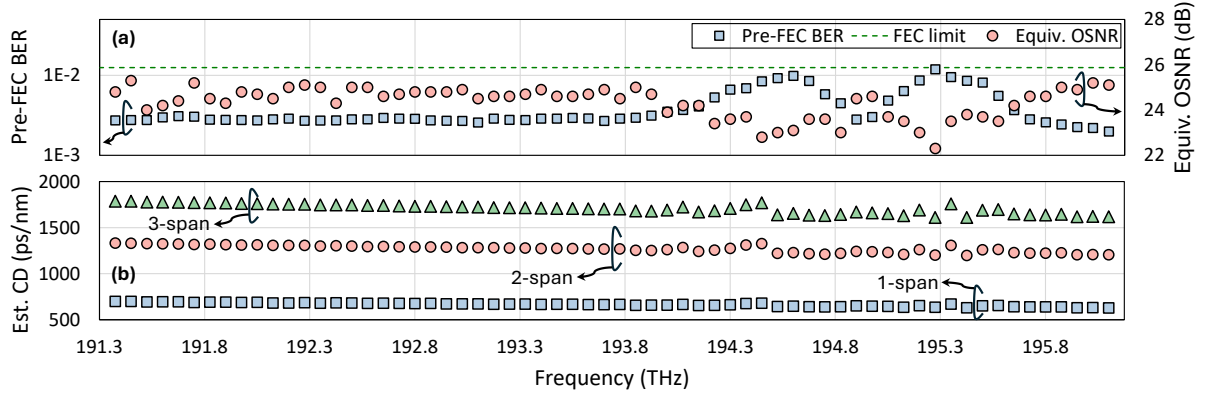


Fig. 3: Results of full C-band 400G ZR transmission over the 3-span HCF link: (a) pre-FEC BER and estimated equivalent OSNR, and (b) estimated accumulated CD values of the HCF links.

case ①) and 2-span 315.97-km (i.e., case ②) HCF links. The pre-FEC BER and estimated equivalent OSNR results are given in Fig. 2 (a) and (b), respectively. After 1-span 163.9-km HCF transmission, pre-FEC BERs slightly above 1×10^{-3} were achieved at all 64 channels, and the estimated equivalent OSNRs were around 27 dB. In addition, the GLA impact after 1-span transmission was negligible, as only minor performance degradation was observed at frequencies around 194.525 THz and 195.275 THz.

After extending the 400G ZR transmission over a 2-span 315.97-km HCF link, the pre-FEC BER performance slightly increased to $\sim 2.1 \times 10^{-3}$ for GLA-free channels, as shown in Fig. 2(a). However, the accumulated GLA impact became more significant after the second span, which we note, did contain a number of fibers with high GLA coefficient. As a result, the pre-FEC BERs at GLA channels exhibited more significant degradations. Nevertheless, all the 64 channels achieved pre-FEC BERs that are below the FEC limit of 1.25×10^{-2} . Furthermore, since the equivalent OSNR was estimated based on the pre-FEC BER, notable reductions were experienced at the GLA channels after 2-span 315.97-km HCF transmission, as expected. In comparison, at GLA-free channels, the estimated equivalent OSNRs only showed minor reductions after two spans, as shown in Fig. 2(b).

Finally, we incorporated the third span to investigate the reach limit of full C-band 400G ZR transmission in our setup. As shown in Fig. 1(a), a 112-km span was adopted to form a 427.97-km HCF link. The corresponding transmission performance is presented in Fig. 3(a). It can be seen that most of the channels (i.e., GLA-free ones) still achieved pre-FEC BERs well below the FEC limit of 1.25×10^{-2} . While GLA imposes performance degradation on the channels around 194.525 THz and 195.275 THz, all these channels however still exhibited pre-FEC BERs below the targeted FEC limit. The corresponding

estimated equivalent OSNRs after 3-span 427.97-km HCF transmission are also given in Fig. 3(a), which again shows considerable reductions due to the GLA impact.

Fig. 3(b) shows the estimated accumulated CD of the three HCF links. After 1-span 163.9-km transmission, the total CD ranged from ~ 630 to ~ 700 ps/nm across the C-band. This further increased to $\sim 1,620$ to $1,790$ ps/nm after 3-span 427.97-km, which is still well below the CD limit of 2,400 ps/nm according to OIF agreement [7]. Therefore, it appears that the impact of GLA rather than CD became the limiting factor to even longer reaches using 400G ZR optics. It is also worth noting that while the transmitter power optimization scheme was proven to be effective to combat the GLA impact for long-distance HCF transmission [15], the optical power tunability range of 400G ZR pluggable is relatively limited, hence the scheme is not explored in this work. We note that the GLA-free channels showed considerable margin to the pre-FEC BER limit, indicating that longer reach could be achieved by using HCFs with low GLA levels. Nevertheless, these results clearly show the potential of HCFs to enable multi-span long-distance links with reduced cost 400G ZR transceiver hardware.

Conclusions

In this paper, we demonstrate record-long reaches of 400G ZR transmission by exploiting the ultra-low nonlinearity and low-and-flat CD of HCF. Full C-band 400G ZR transmission is achieved over up to 3-span 427.97-km HCF links, whilst the pre-FEC BERs of all 64 channels across the C-band were below the FEC limit of 1.25×10^{-2} with gas line absorption being the limit for the worst affected channels. These results provide new insight into using 400G ZR optics for optical interconnects beyond metro distances, which provides cost and energy efficiency gains as well as operational simplicity for future network implementations.

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